EXP 3

Editor

ADCexp2.m

/MATLAB Drive/ADCexp2.m

```
clc;
clear;
close all;

Am = 1;
fm = 500;
Ac = 5;
fc = 5000;

fs = 20*fc;
t = 0:1/fs:4/fm;

% Message signal
m = Am*sin(2*pi*fm*t);

% Carrier signal
c = Ac*sin(2*pi*fc*t);

% Modulation index
mu = Am/Ac;

% AM signal
am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

```

```
% Demodulation using Hilbert transform
dem = abs(hilbert(am));
dem = dem - mean(dem);

% FFT
N = length(am);
f = linspace(-fs/2, fs/2, N);
AM_FFT = abs(fftshift(fft(am)))/N;

% Sideband frequencies
USB = fc + fm;
LSB = fc - fm;

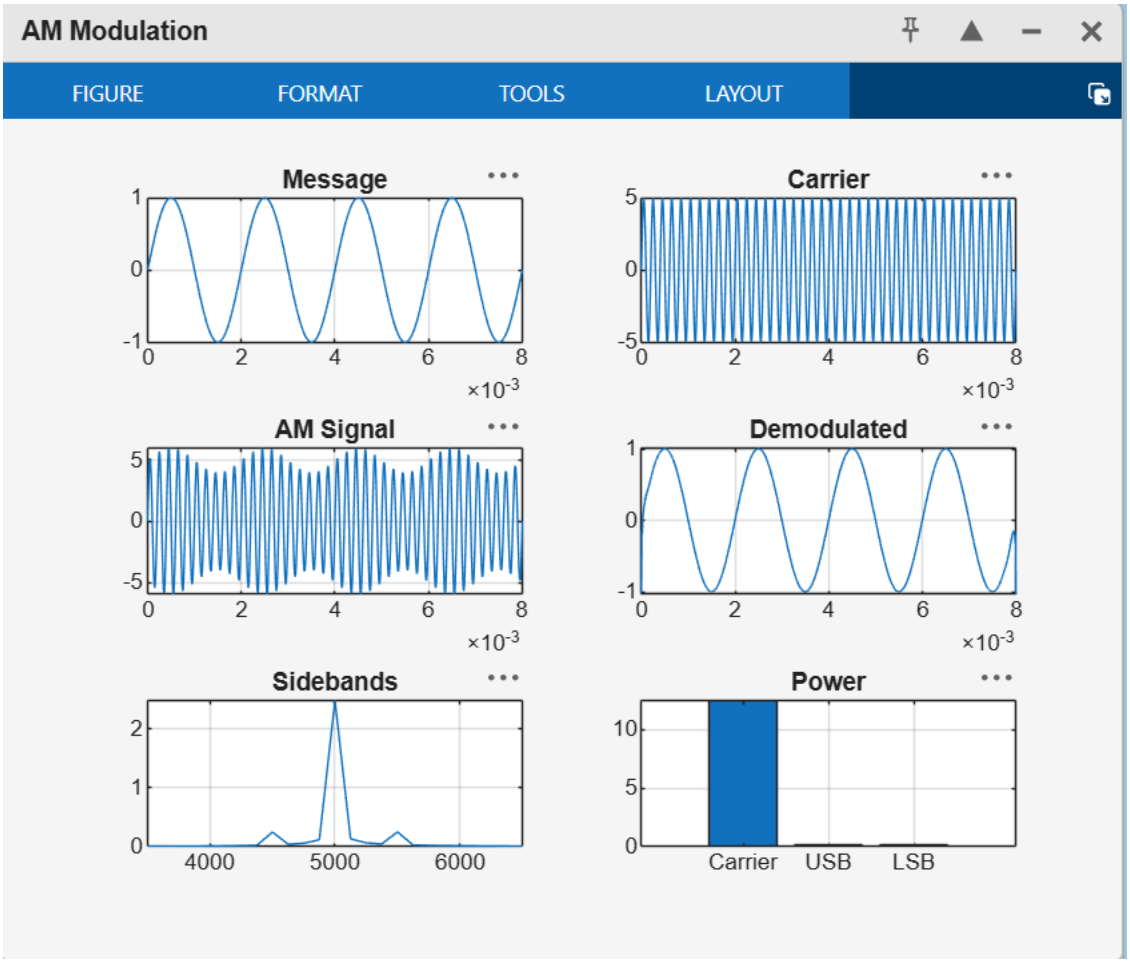
% Power calculations
Pc = Ac^2/2;
Psb = (mu^2 * Pc)/4;

% Plotting
figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1);
plot(t,m);
title('Message');
grid on;
```

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```
50      subplot(3,2,2);
51      plot(t,c);
52      title('Carrier');
53      grid on;
54
55      subplot(3,2,3);
56      plot(t,am);
57      title('AM Signal');
58      grid on;
59
60      subplot(3,2,4);
61      plot(t,dem);
62      title('Demodulated');
63      grid on;
64
65      subplot(3,2,5);
66      plot(f,AM_FFT);
67      xlim([fc-3*fm fc+3*fm]);
68      title('Sidebands');
69      grid on;
70
71      subplot(3,2,6);
72      bar([Pc Psb Psb]);
73      set(gca,'XTickLabel',{'Carrier','USB','LSB'});
```


AM Modulation
FIGURE
FORMAT
TOOLS
LAYOUT
Message
Carrier
AM Signal
Demodulated
Sidebands
Power
×10-3
×10-3
×10-3
×10-3
4000
5000
6000
Carrier
USB
LSB